7 **(a)** **(i)** State what is meant by half-wave rectification of an alternating current.

....

....

.... [2]

(ii) Figure 7.1 shows an incomplete diagram of the circuit used to achieve half-wave rectification of an alternating input voltage V_{IN} to produce output voltage V_{OUT}.

Figure 7.1

Complete the circuit diagram in Figure 7.1. [2]

(b) Two alternating voltages V_1 and V_2 have the same peak voltage V_0 and the same period T. One is sinusoidal and the other is a square wave, as shown by the variations with time t in Figure 7.2 and Figure 7.3.

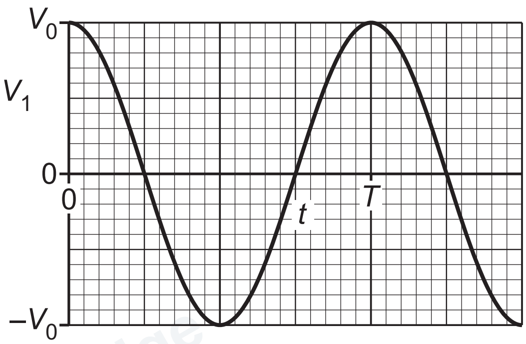

Figure 7.2

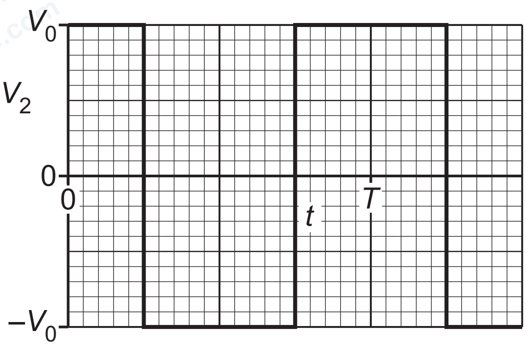

Figure 7.3

The alternating voltages are applied in turn to the input terminals in the completed circuit of Figure 7.1.

Complete Table 7.1 to show, in terms of V_0, expressions for the root-mean-square (r.m.s.) values of the two unrectified and the two half-wave rectified voltages.

Table 7.1

	r.m.s. value	
	$\boldsymbol{V_1}$	$\boldsymbol{V_2}$
unrectified		
half-wave rectified		

[4]

[Total: 8]